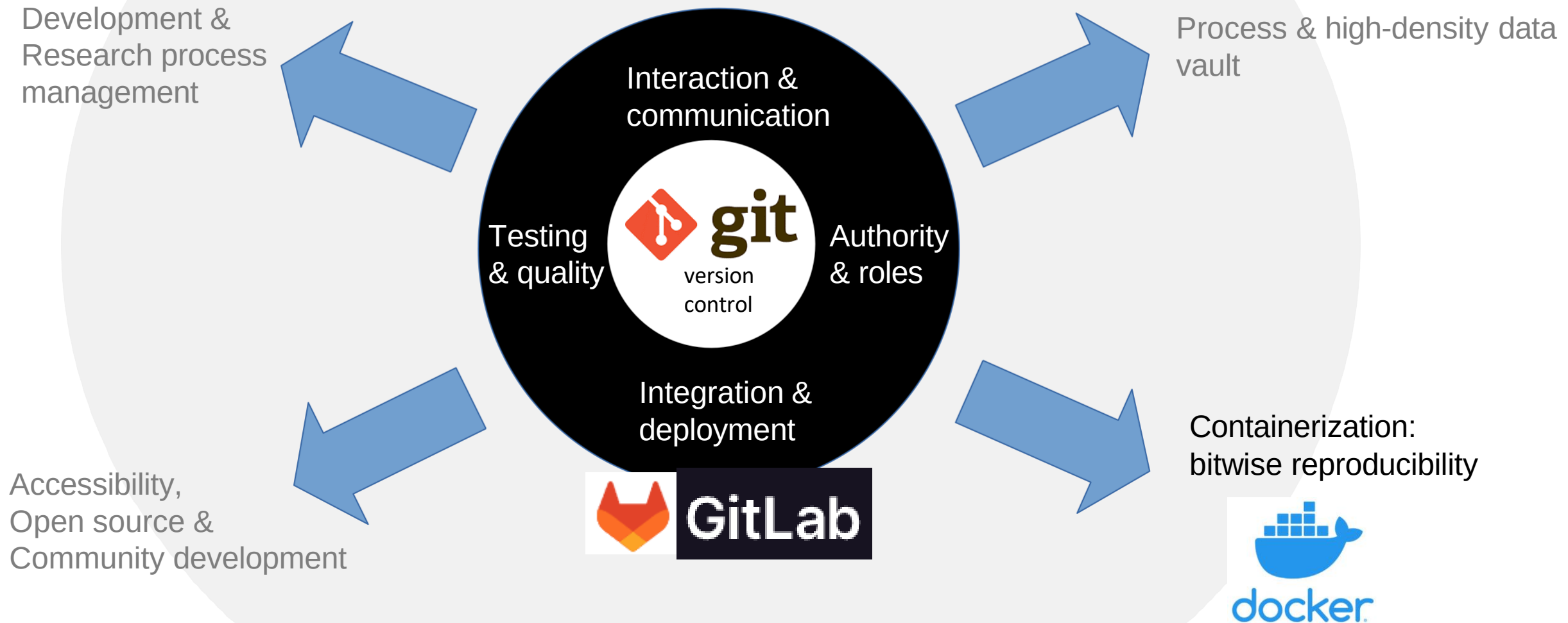
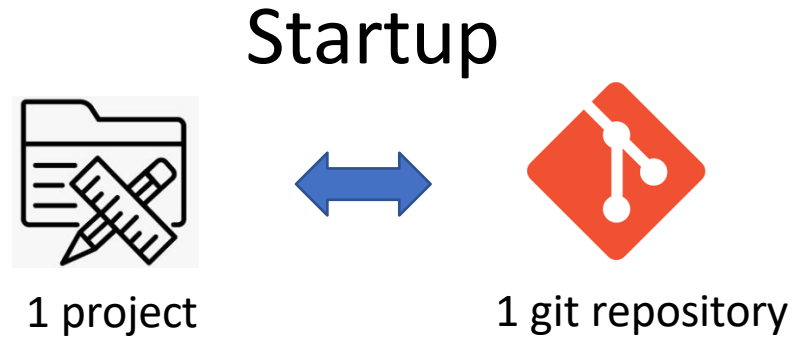


Coding: collaborative,
trustworthy, reproducible

An infrastructure in shells



Version control with Git



Linus Torvalds, 2005
(frustrated by bitkeeper)



Repo examples:

- 1 code project
- 1 paper
- 1 student
- 1 project proposal
- 1 presentation

```
cd project_folder  
git init .
```

Next slides: 8 basic commands

Git: basic version control

(1 developer)

- git init
- git add stage file for snapshot
- git commit take snapshot
- git diff compare snapshots
- git log snapshot list
- git checkout move to a different snapshot
- git branch branch the future
- git merge join different futures in one

Tracking versions

Make a snapshot of the current version

 \$ git **add** file1.py file2.py

 \$ git **commit** -m 'added files'

And so on for the next snapshot

 \$ git add file1.py

 \$ git commit -m 'changed XX YY'

Changes

git **diff**

```
my_code_2
[acorbe@Alessandros-MacBook-Pro my_code_2 [master] $ git diff
diff --git a/converter.py b/converter.py
index 4511134..58edb46 100644
--- a/converter.py
+++ b/converter.py
@@ -1,13 +1,16 @@
+from __future__ import print_function
import numpy as np
+import sys
+
def binary_number_string_parser(inp
    , datatype = float):

-
+    print (inp)
    return 0

def main():
-
-    return binary_number_string_parser(0)
+
+    return binary_number_string_parser(sys.argv[1])

if __name__ == '__main__':
[acorbe@Alessandros-MacBook-Pro my_code_2 [master] $
```

Removed content

Added content

Changes

git log

```
my_code_2 —
[acorbe@Alessandros-MacBook-Pro my_code_2 [master] $ git add conv
[acorbe@Alessandros-MacBook-Pro my_code_2 [master] $ git commit -r
[master 96bf846] added command line capturing
 1 file changed, 6 insertions(+), 3 deletions(-)
[acorbe@Alessandros-MacBook-Pro my_code_2 [master] $ git log
commit 96bf846d46db1177cd04613cbb9dbdd71dd1d6f7 (HEAD -> master)
Author: Alessandro Corbetta <corbisoft@gmail.com>
Date: Tue May 1 16:39:02 2018 +0430

    added command line capturing

commit ac1408245b78a7a8b53f5040b8780c1733fdf6af
Author: Alessandro Corbetta <corbisoft@gmail.com>
Date: Tue May 1 16:27:09 2018 +0430

    initial commit
[acorbe@Alessandros-MacBook-Pro my_code_2 [master] $
```

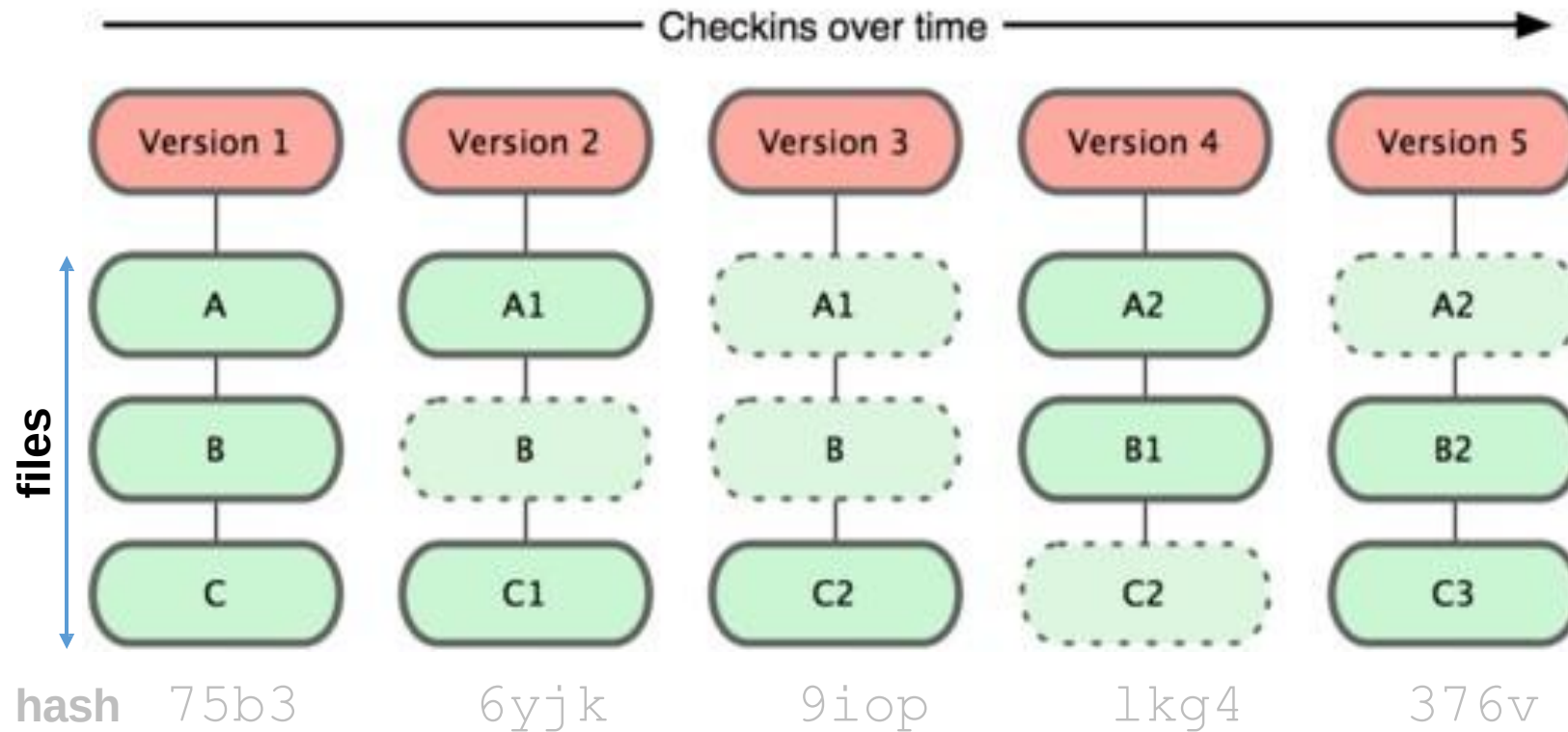
Second snapshot (commit)

First snapshot (commit)

History & rollback

- Each commit holds a complete snapshot of the repository

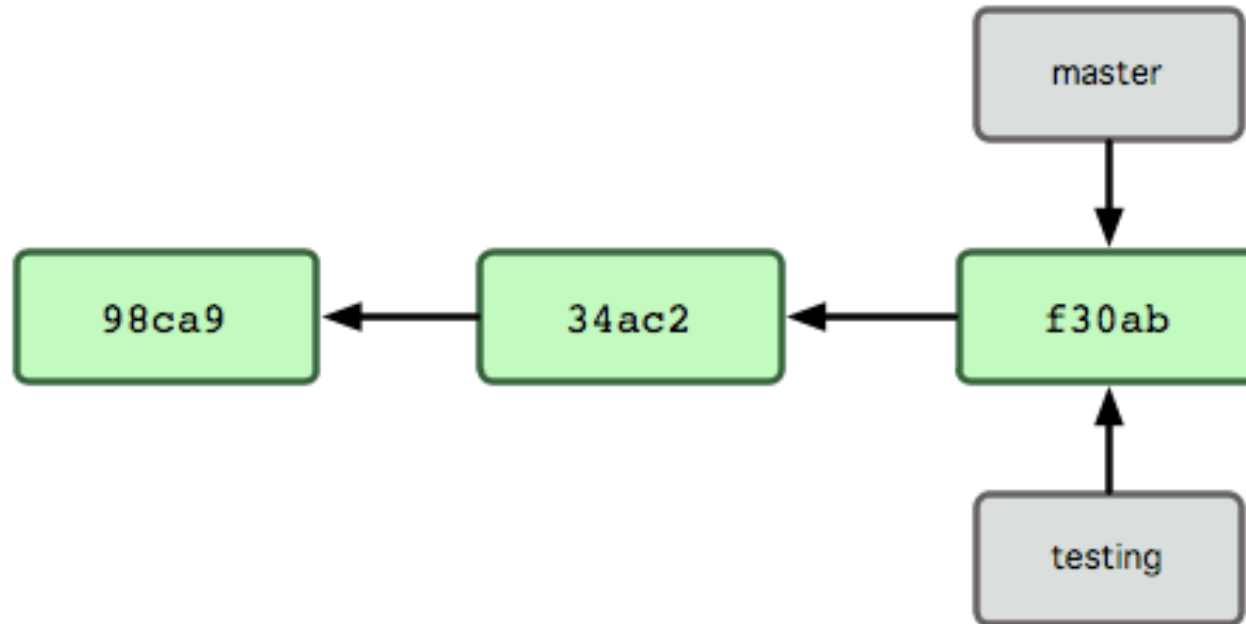
`git checkout <hash>`



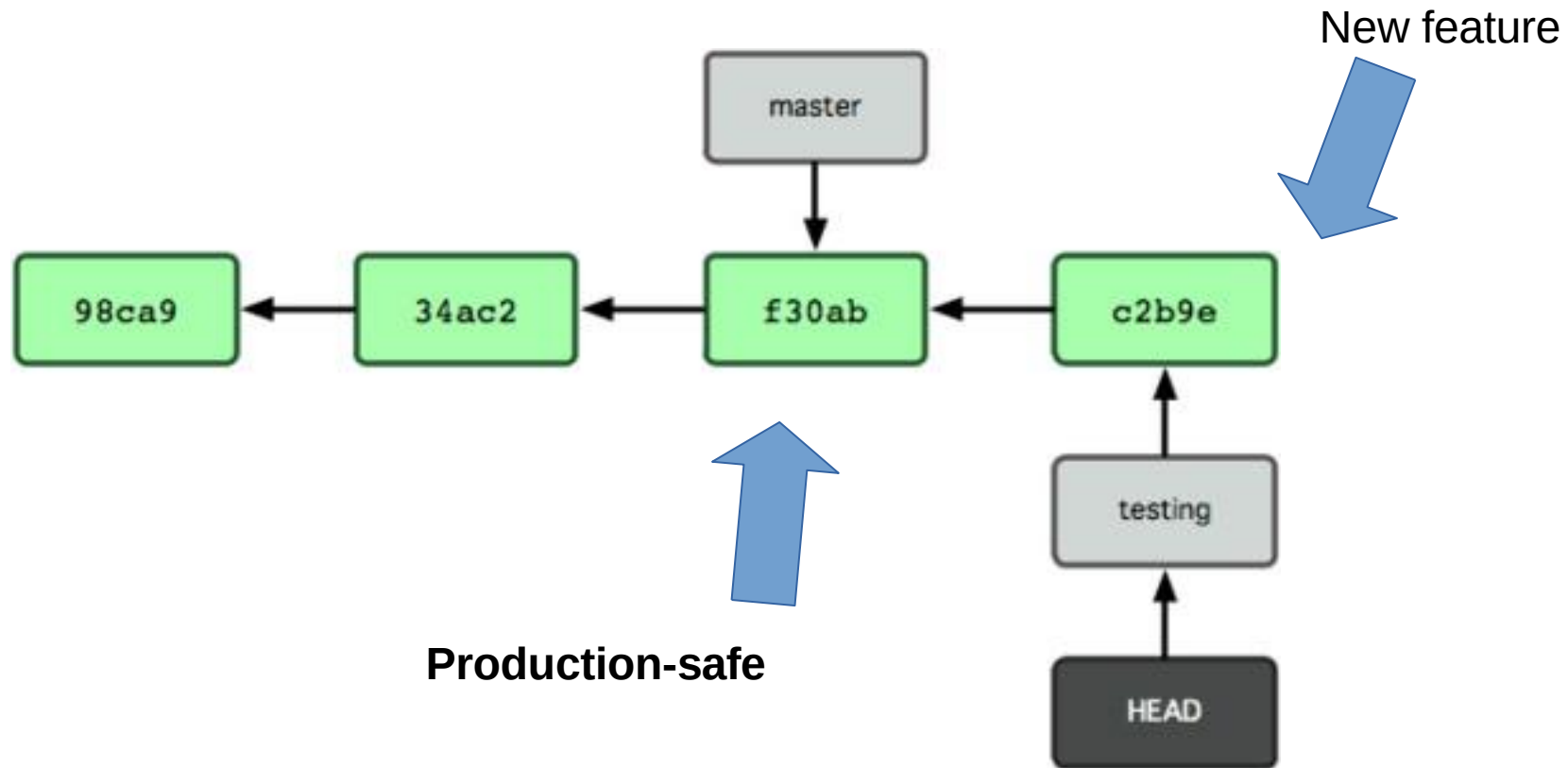
Branches for a safe future

Default branch “master” or “main”

New branch: `git branch testing`

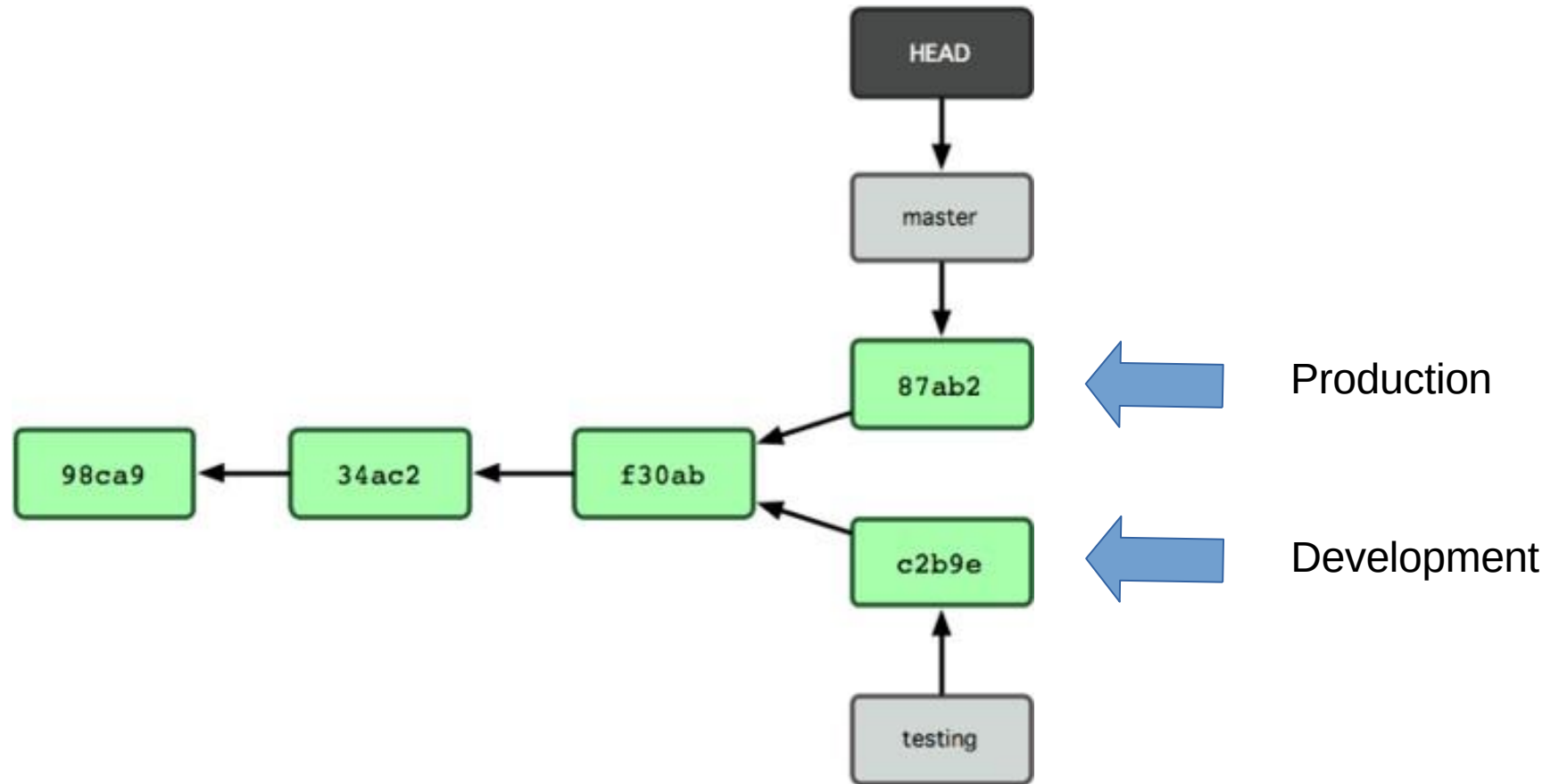


Development in “testing”



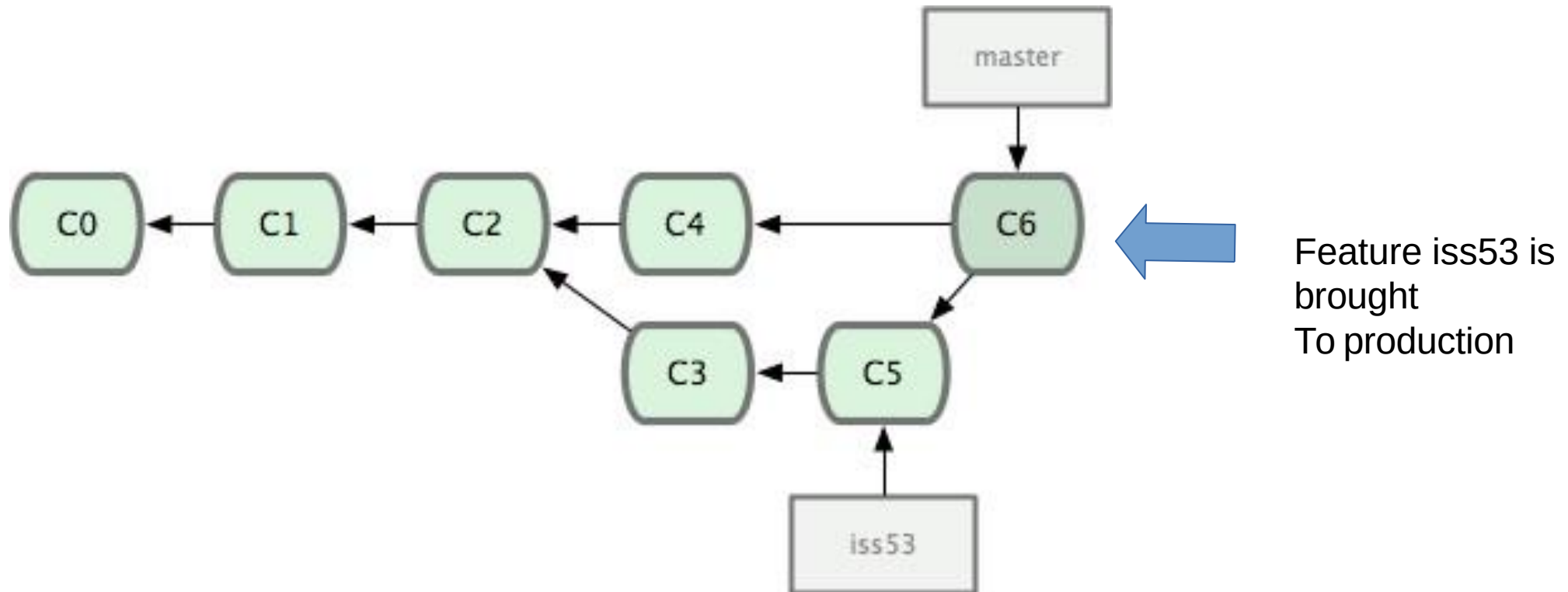
Branches can evolve independently!

(typically is not what you wanna do, master is only one!)



Merging (to the the only master 😊)

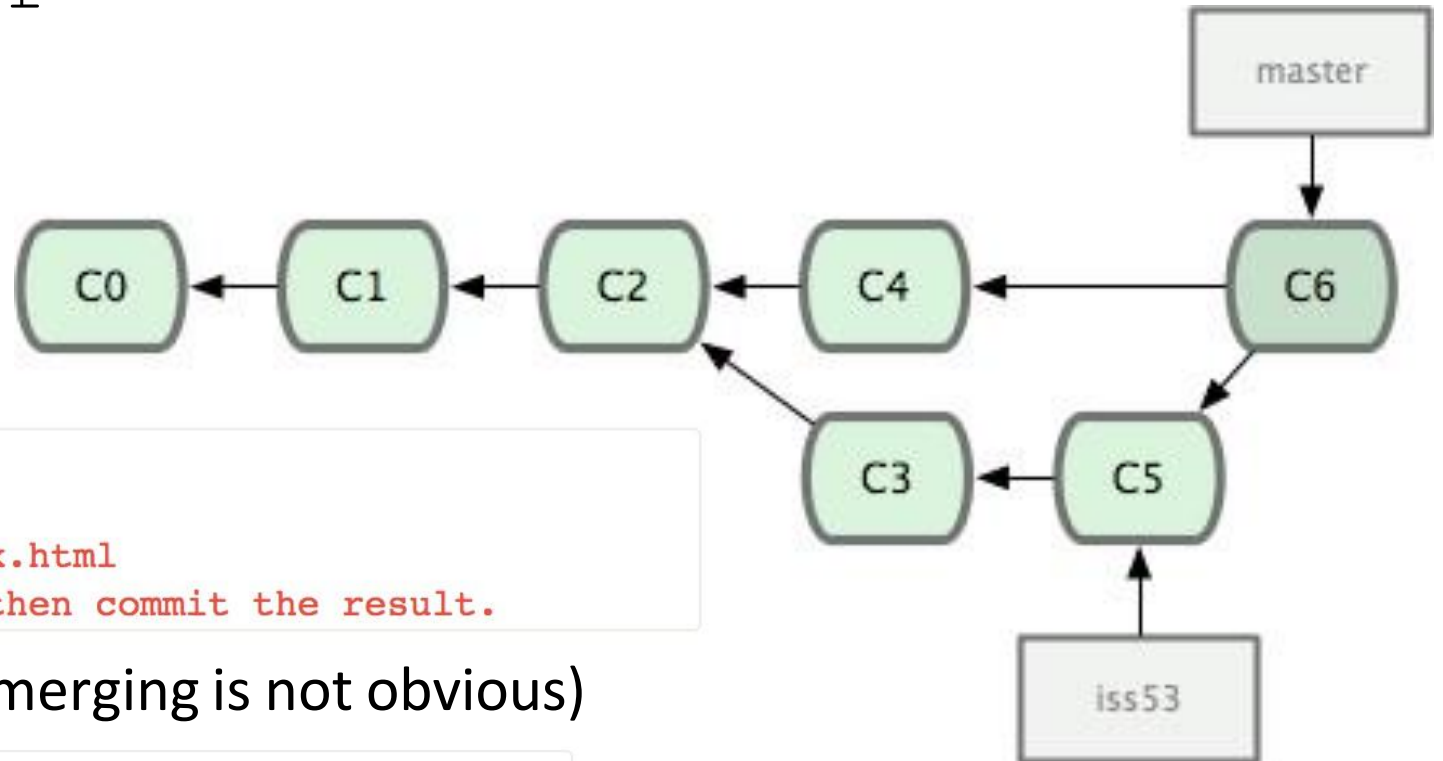
- `git checkout master`
- `git merge iss53`



Merge **conflicts**

certain decisions are for humans only

- git checkout master
- git **merge** iss53



```
$ git merge iss53
Auto-merging index.html
CONFLICT (content): Merge conflict in index.html
Automatic merge failed; fix conflicts and then commit the result.
```

Manual merging might be needed (if merging is not obvious)

```
<<<<<< HEAD:index.html
<div id="footer">contact : email.support@github.com</div>
=====
<div id="footer">
  please contact us at support@github.com
</div>
>>>>>> iss53:index.html
```

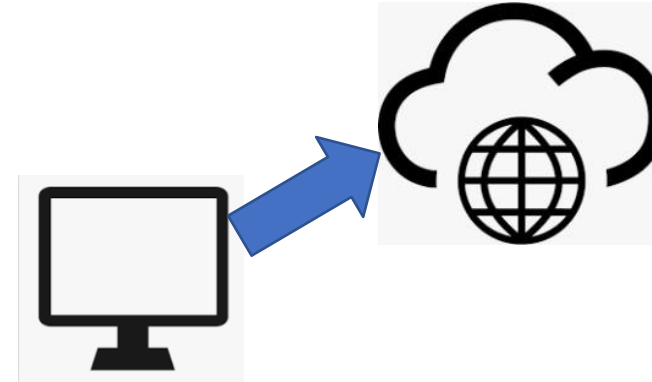
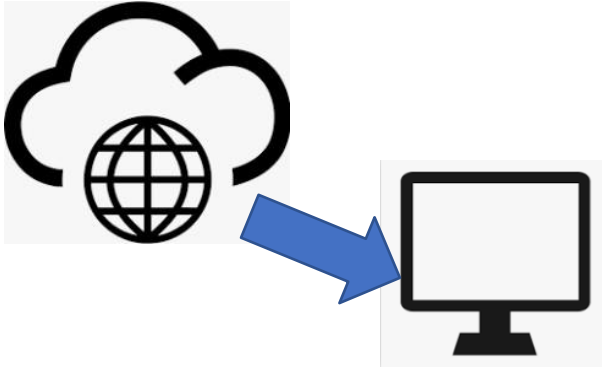
Git: basic version control

(1 developer)

- git init
- git add stage file for snapshot
- git commit take snapshot
- git diff compare snapshots
- git log snapshot list
- git checkout move to a different snapshot
- git branch branch the future
- git merge join different futures in one

Coding: collaborative,
trustworthy, reproducible

Pulling/Pushing commits to the origin



To update <branch> locally

- `git checkout <branch>`
- `git pull origin <branch>`

To push to remote

- `git checkout <branch>`
- `git push origin <branch>`

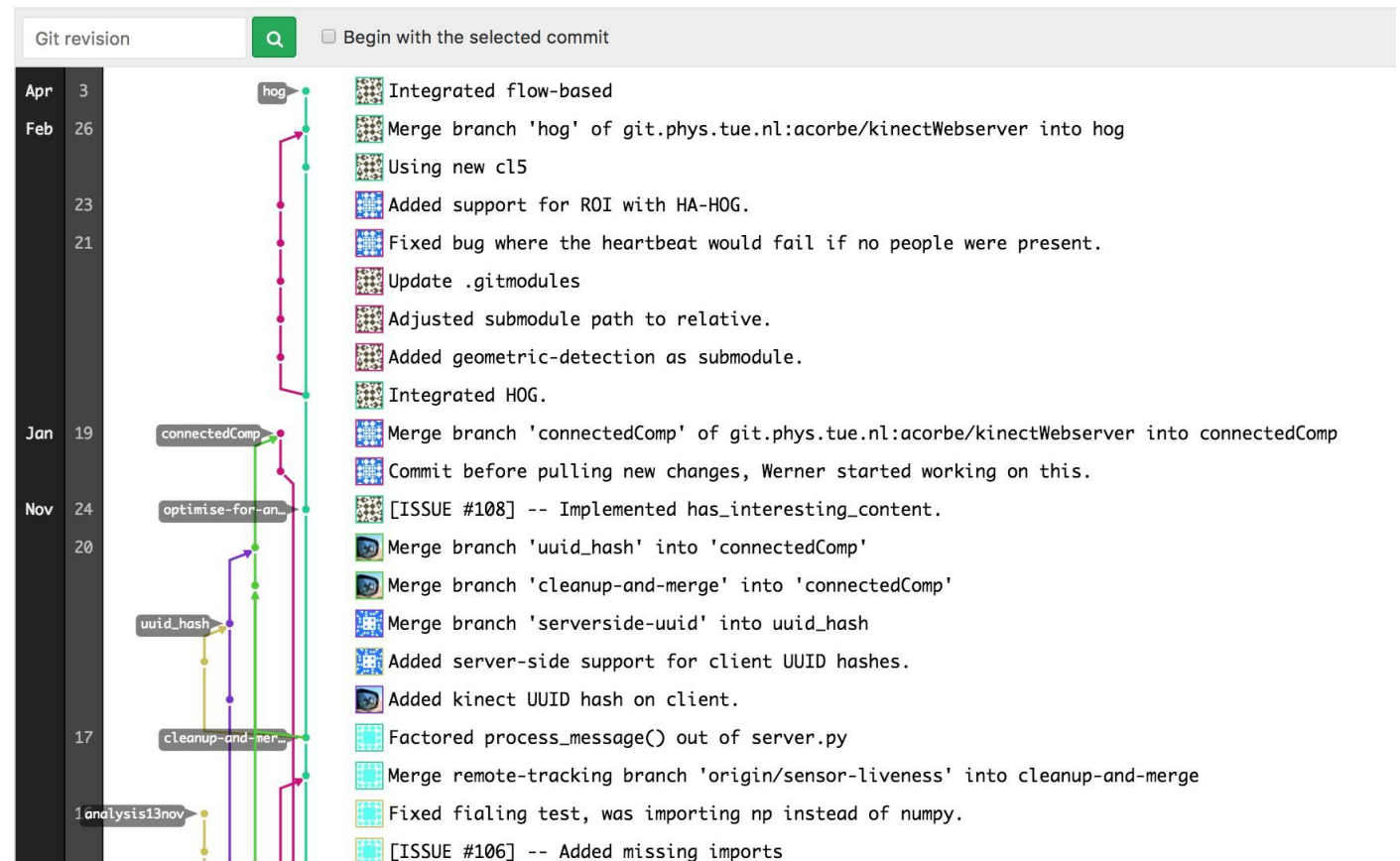
Note: remotes can be **public** or **private**

Pulling/Pushing commits to the origin



Alessandro C > kinectWebserver > Commits

hog	kinectWebserver
03 Apr, 2018 1 commit	
	Integrated flow-based Werner Kroneman authored 4 weeks ago
26 Feb, 2018 2 commits	
	Merge branch 'hog' of git.phys.tue.nl:acorbe/kinectWebserver into hog Werner Kroneman authored 2 months ago
	Using new cl5 Werner Kroneman authored 2 months ago
23 Feb, 2018 1 commit	
	Added support for ROI with HA-HOG. Werner Kroneman authored 2 months ago
21 Feb, 2018 5 commits	
	Fixed bug where the heartbeat would fail if no people were present. Werner Kroneman authored 2 months ago
	Update .gitmodules Werner Kroneman authored 2 months ago
	Adjusted submodule path to relative. Werner Kroneman authored 2 months ago



Interaction between two or more developers

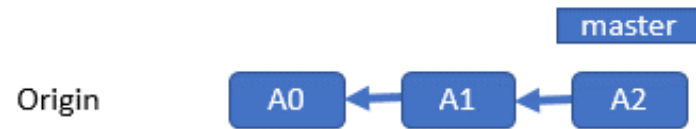
`git clone / push / pull`

collaboration building blocks → How to operate them?

Interaction between two or more developers

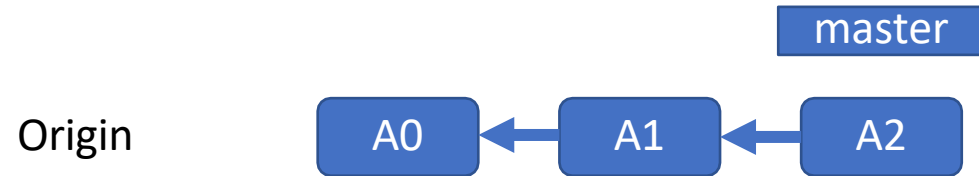
Collaborative development

What happens after clone?



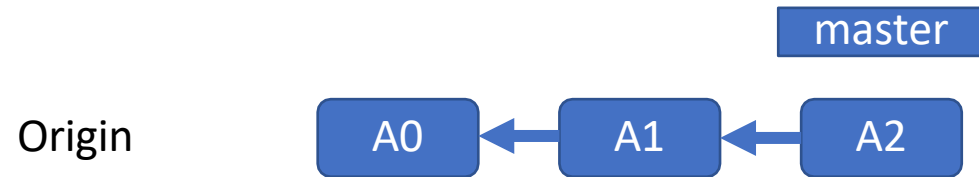
Collaborative development

What happens after clone?



Collaborative development

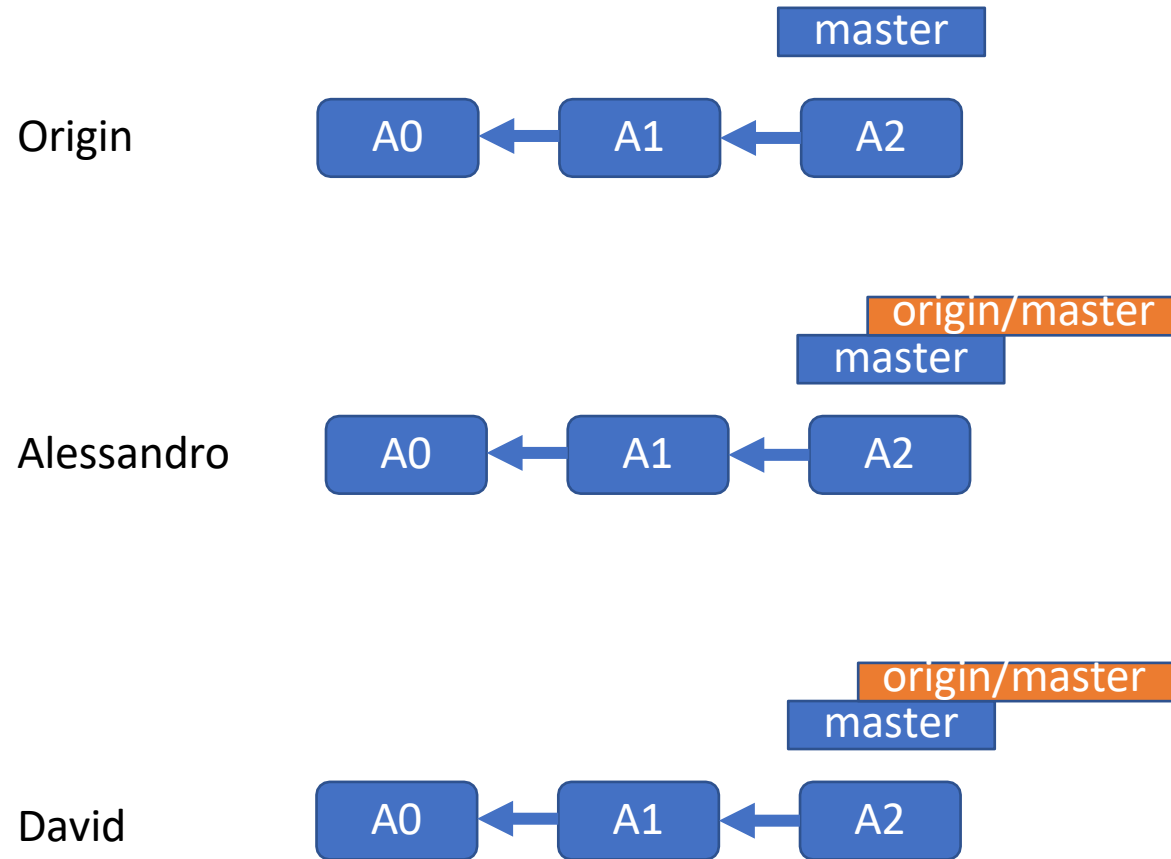
What happens after clone?



Alessandro & David both do `git clone`

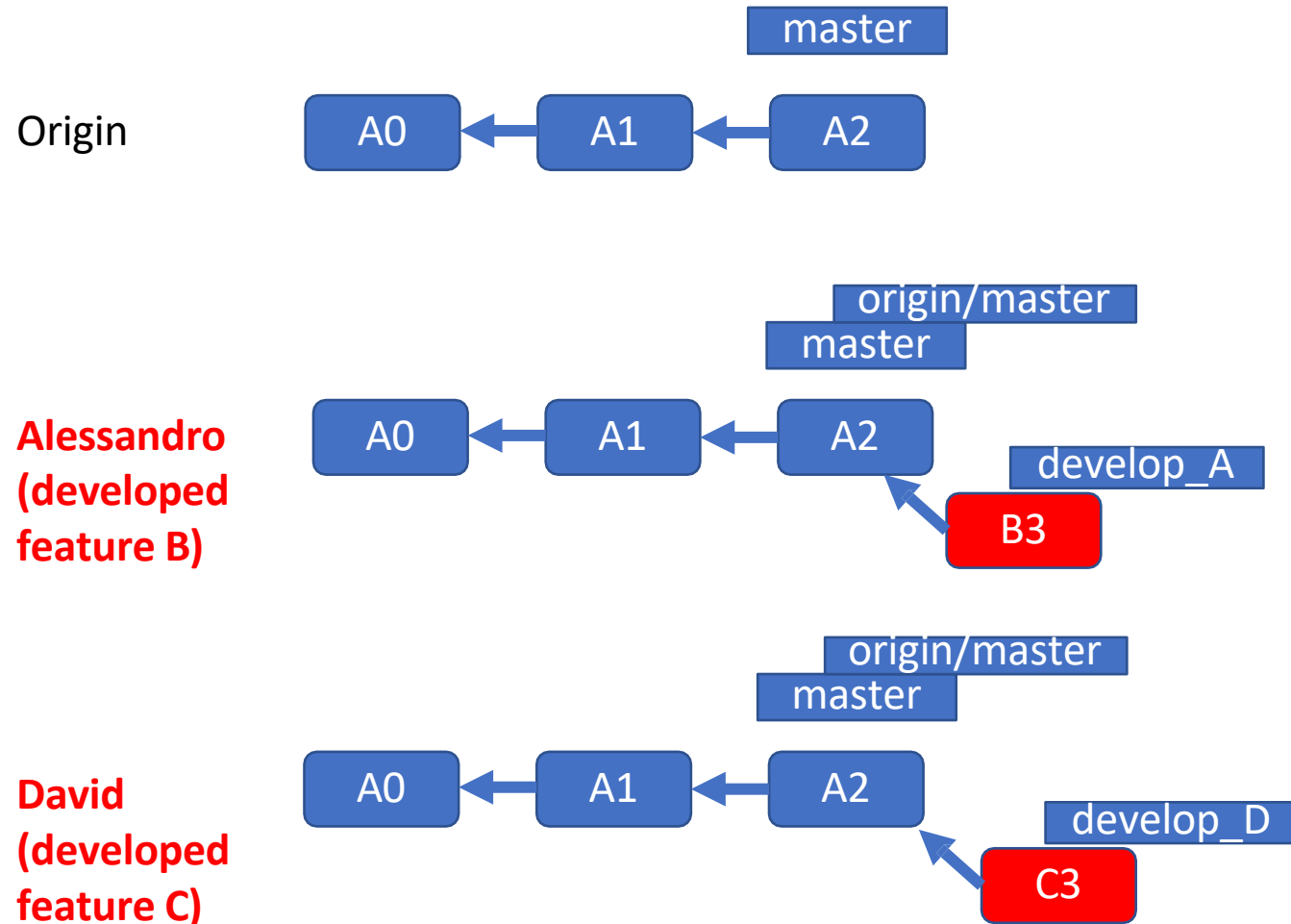
Collaborative development

What happens after clone?



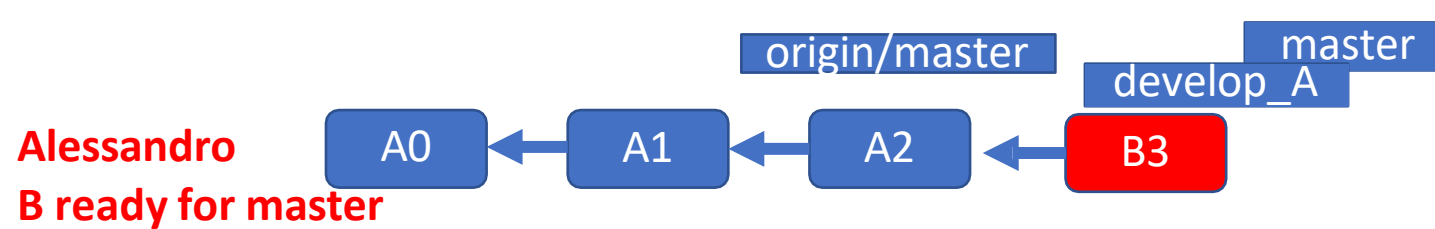
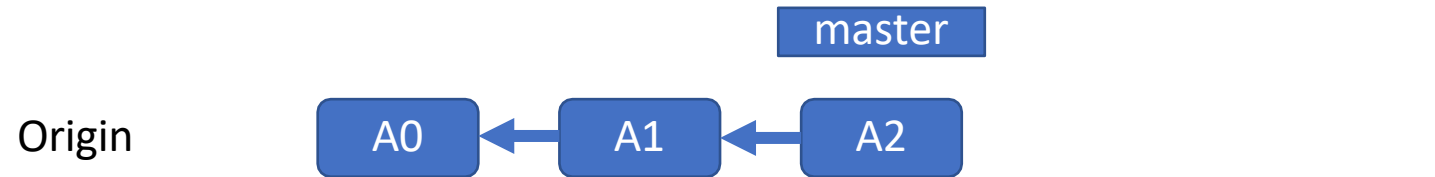
Collaborative development

What happens after clone?

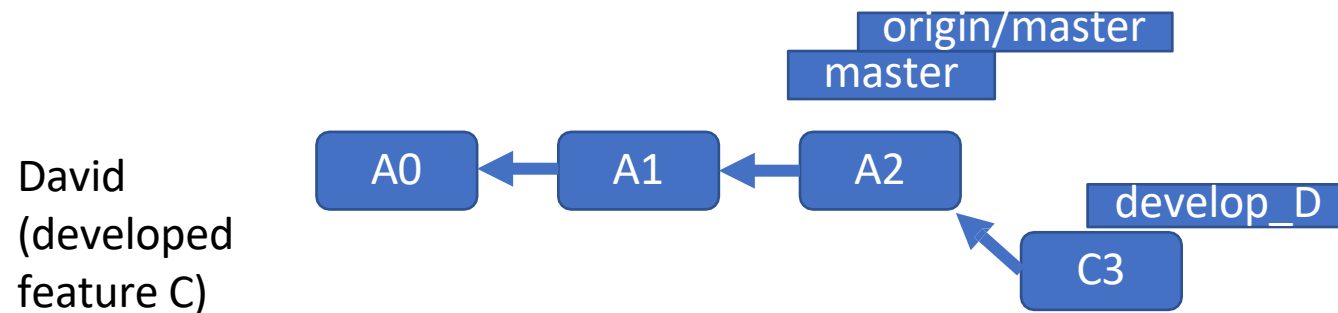


Collaborative development

What happens after clone?

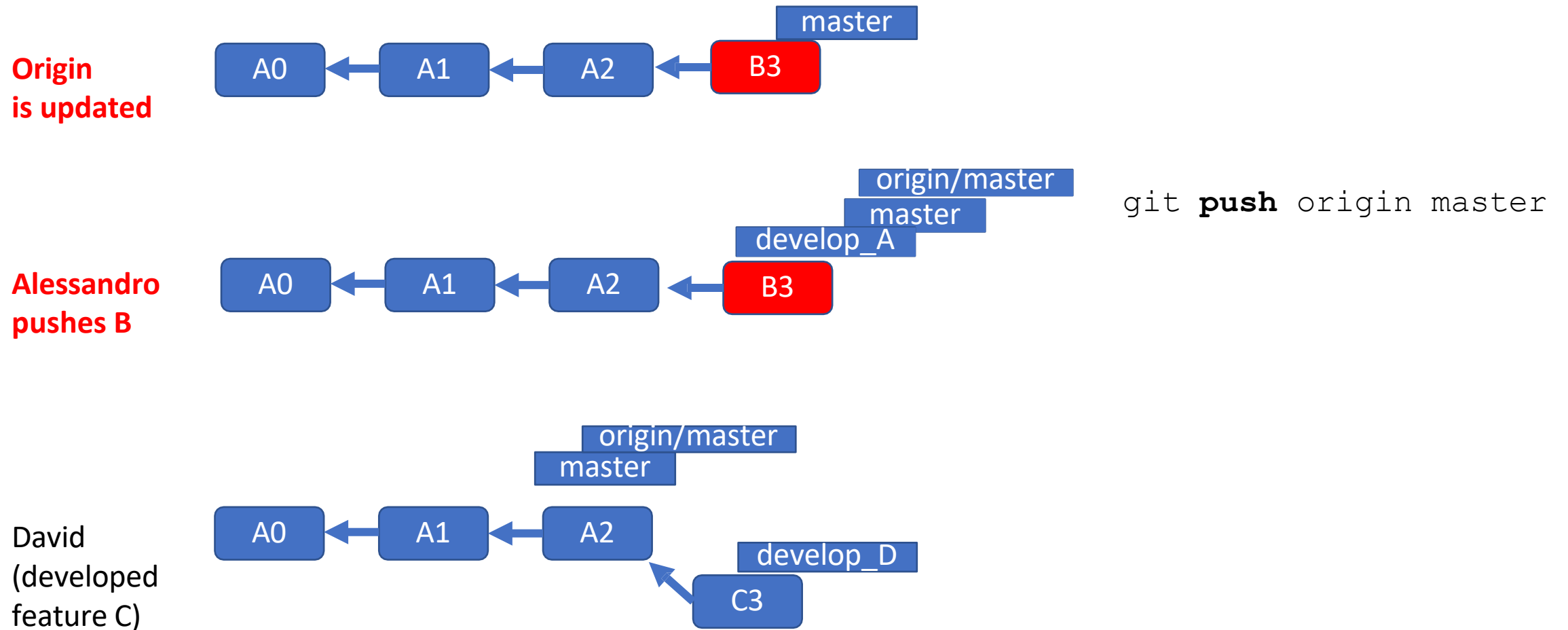


```
git checkout master  
git pull  
git merge develop_A
```



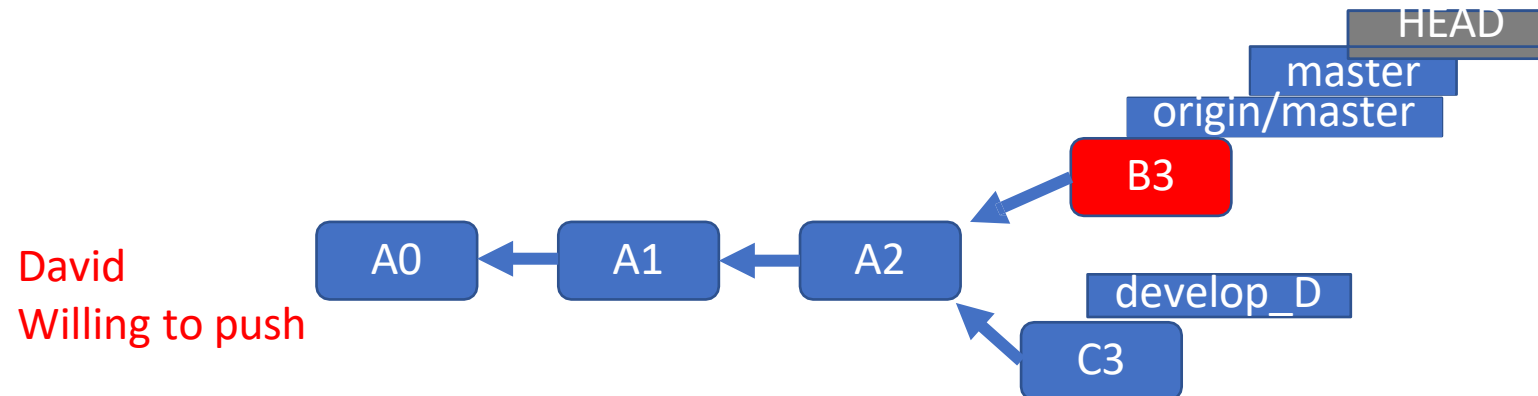
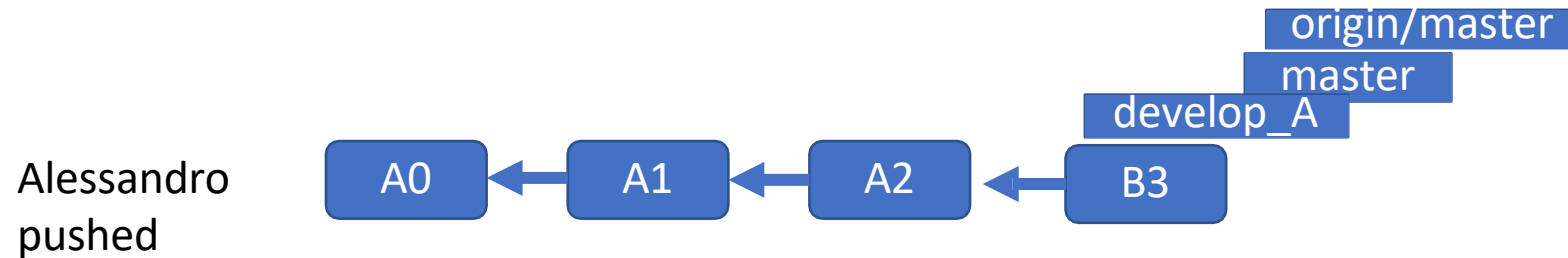
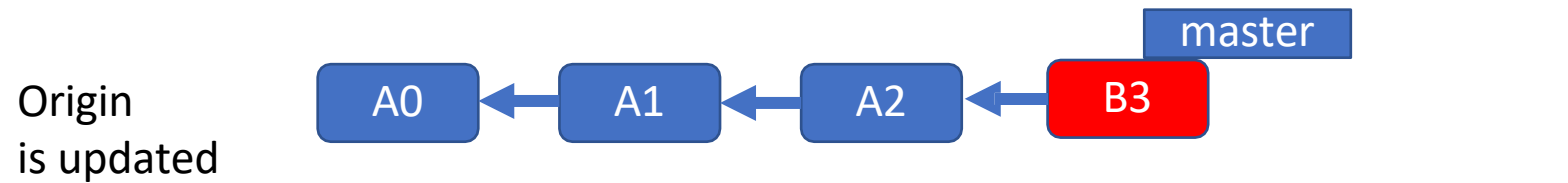
Collaborative development

What happens after clone?



Collaborative development

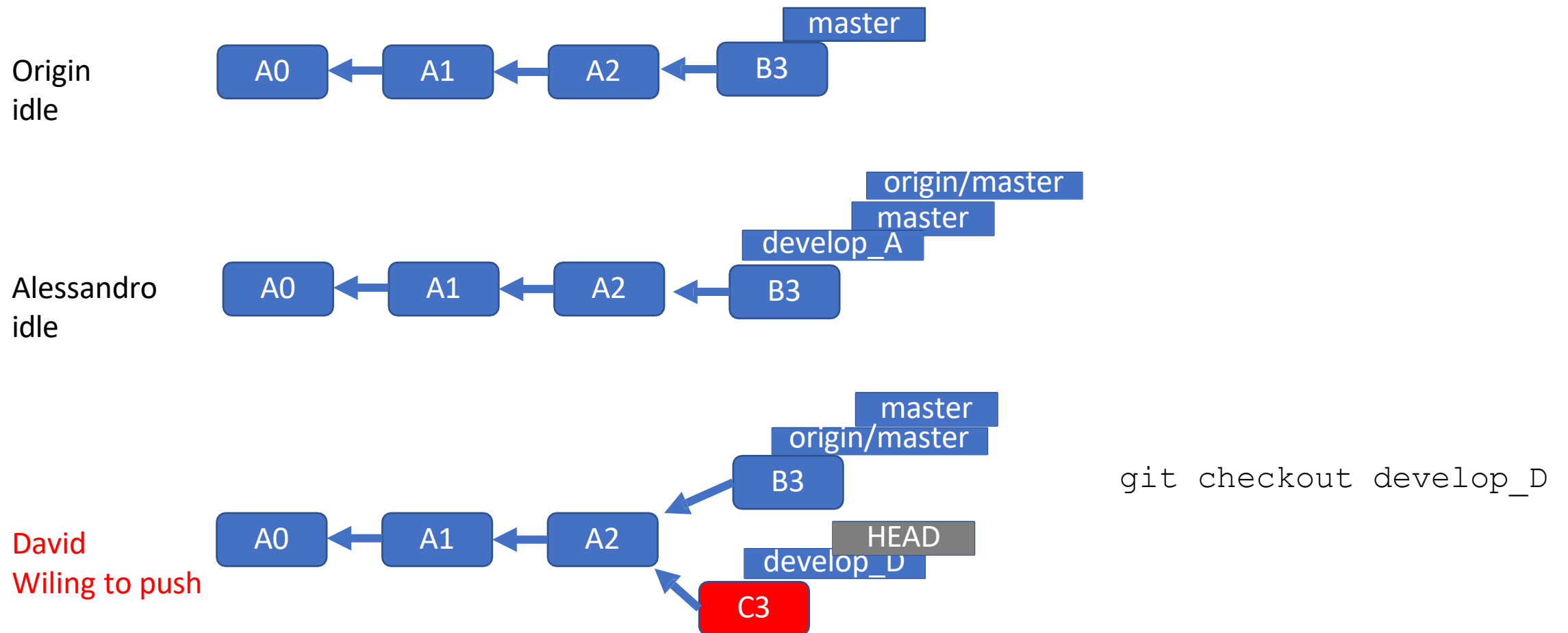
What happens after clone?



```
git checkout master  
git pull #UPDATES!!
```

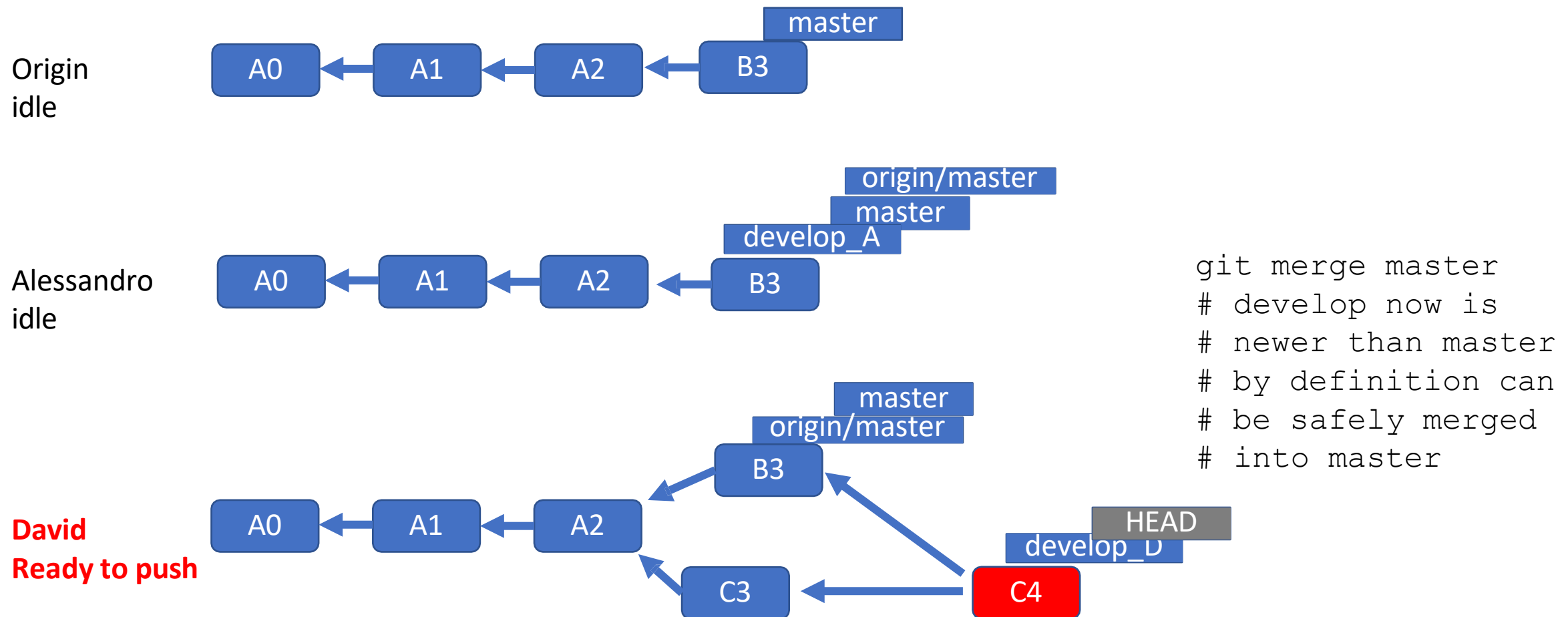
Collaborative development

What happens after clone?



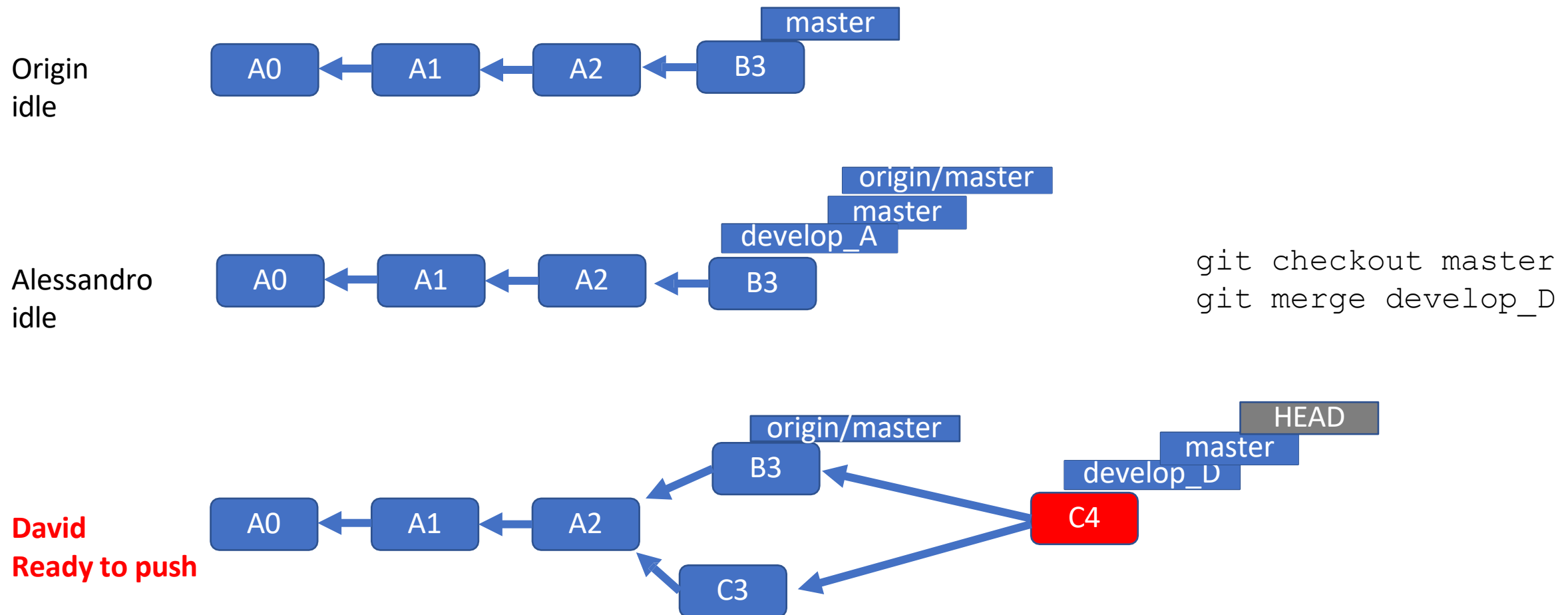
Collaborative development

What happens after clone?



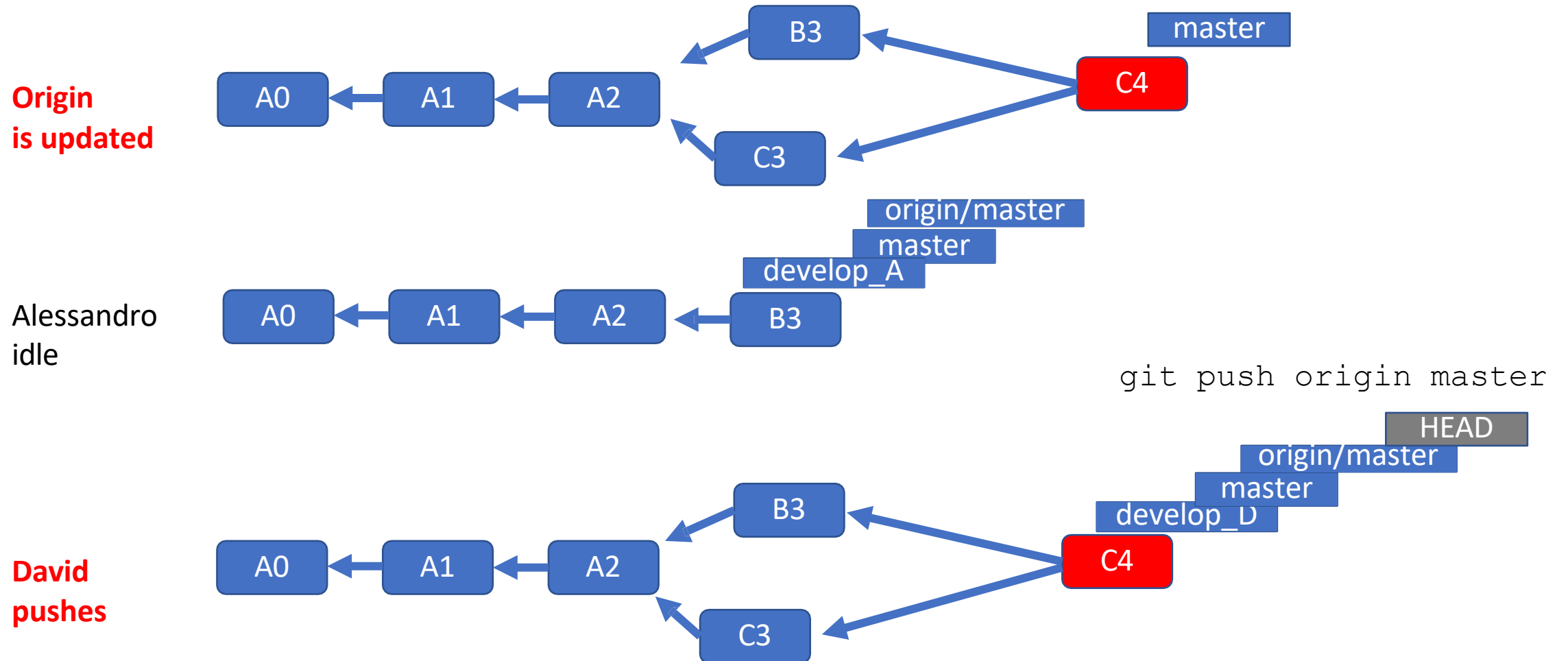
Collaborative development

What happens after clone?



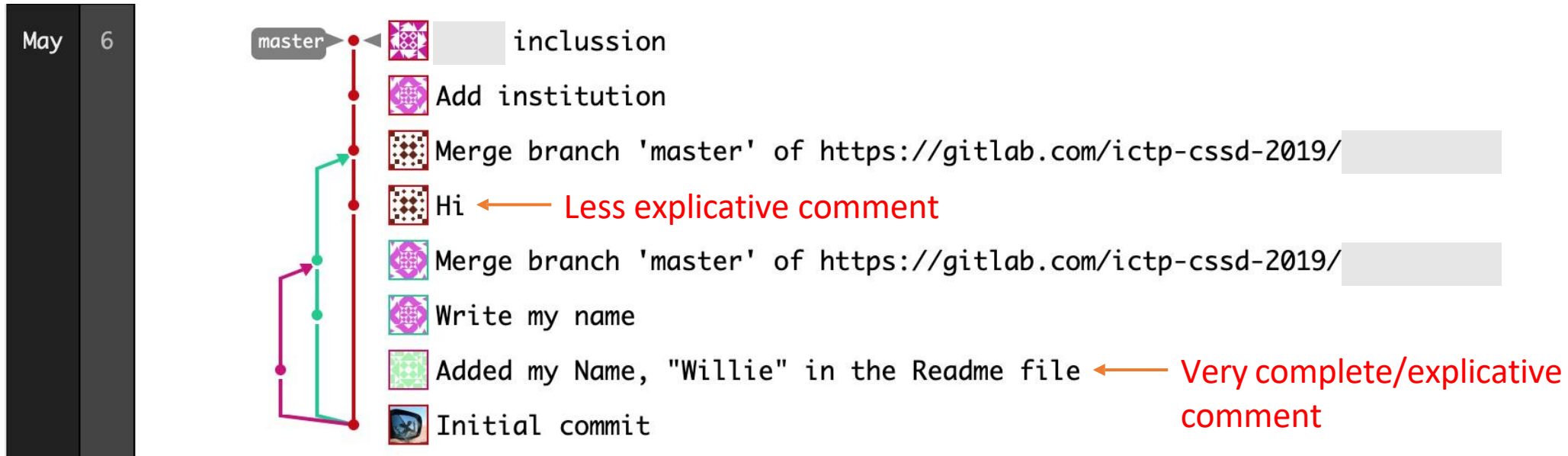
Collaborative development

What happens after clone?



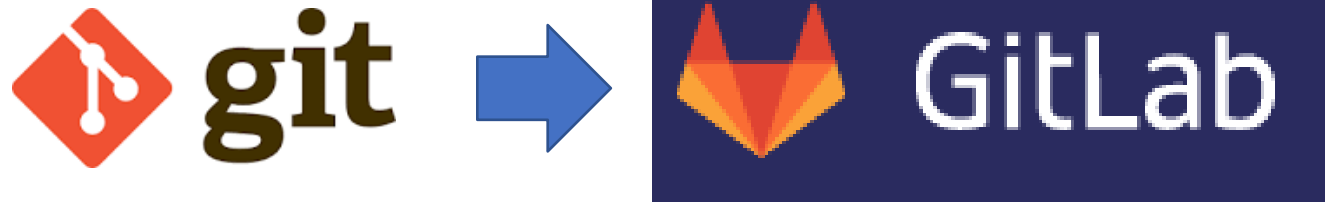
Maximizing scalability and effectiveness

- If all developers push to master, quality can degrade
- How to coordinate?
- How can we trust the changes?

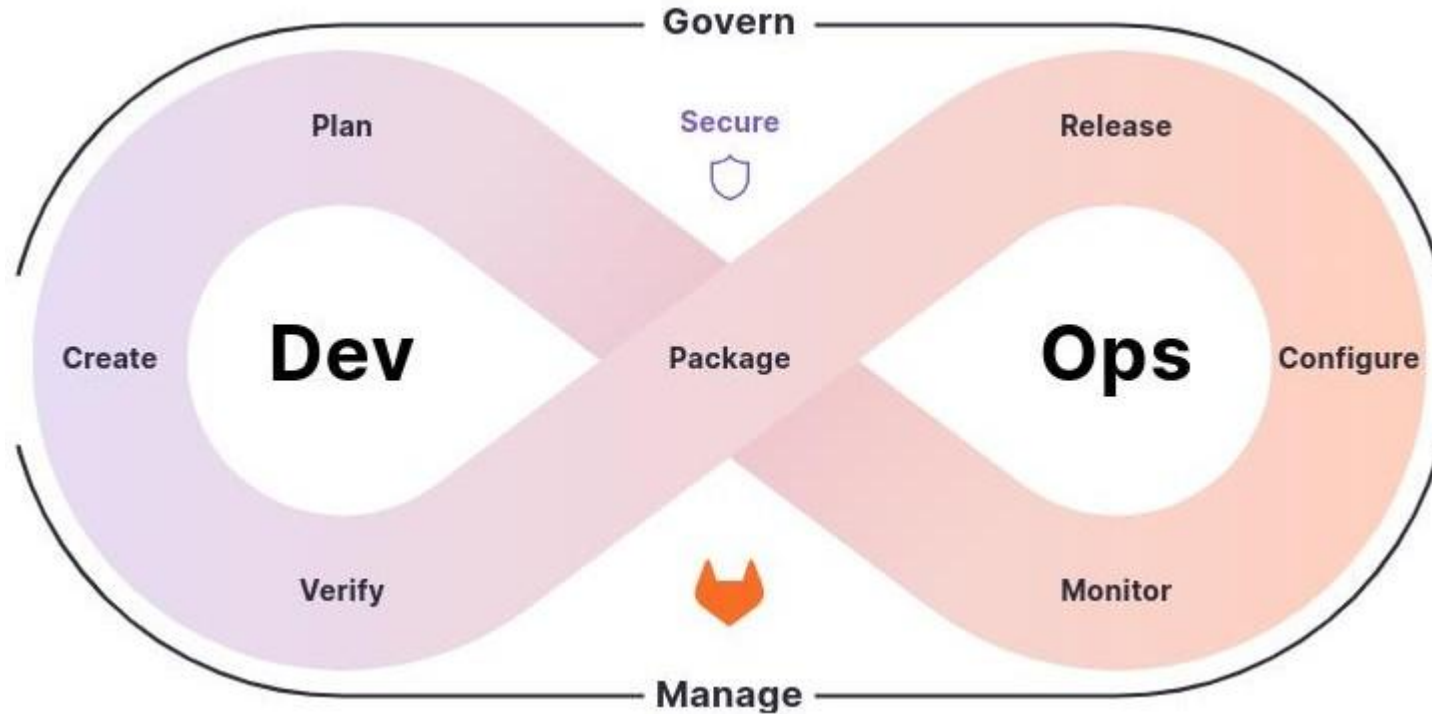


Maximizing scalability and effectiveness?

- If all developers push to master, quality can degrade
- How to coordinate?
- How can we trust the changes?



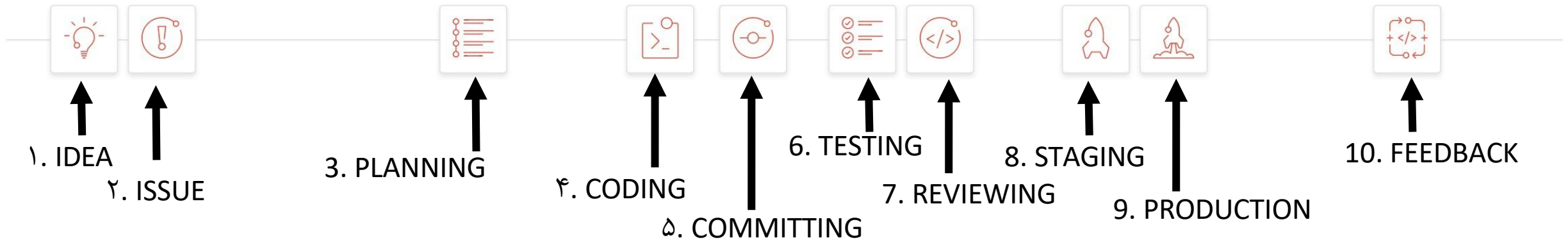
DevOps



DevOps is a combination of software development (dev) & operations (ops).

Software engineering methodology integrating development operations by facilitating collaboration & shared responsibility.

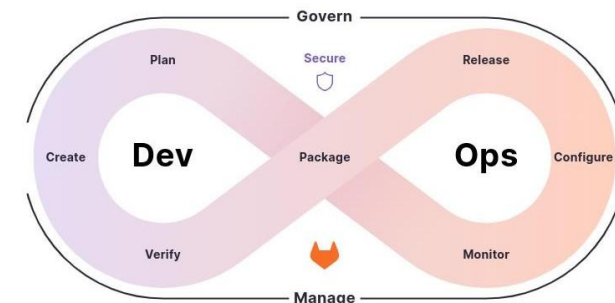
The conversational development paradigm



Gitlab/github: meant to support this approach

Development model => content & conversations between developers
Fosters collaborations w/o centralized entities

In my experience: very scalable also in research (codes & execution)!



Super-structure to a Git repository

The screenshot shows the GitLab interface for the repository 'lbe-direct'. The left sidebar contains navigation links: Overview, Details, Activity, Cycle Analytics, Repository, Issues (132), Merge Requests (0), CI / CD, Registry, Wiki, and Settings. The main content area displays the repository details, including the name 'lbe-direct', the description 'The Lattice Boltzmann code under ftmake', and statistics (3 stars, 0 forks). Below this, there are links to 'Add Changelog', 'Add Contribution guide', and 'Add Kubernetes cluster'. The 'Files' section shows a list of directories and files with their last commit and update dates. The 'Merge branch' section shows a merge of 'develop-ISSUE283' into 'master' by Federico Toschi 2 weeks ago. The 'Repository' section shows a list of files and their last commit and update dates.

Annotations on the right side of the image explain the super-structure:

- What others developer did?** points to the **Activity** link in the sidebar.
- Content/commits** points to the **Repository** link in the sidebar.
- Issues** points to the **Issues** link in the sidebar, which has a badge showing 132 issues.
- Merging your contribution** points to the **Merge Requests** link in the sidebar, which has a badge showing 0 merge requests.
- Testing** points to the **CI / CD** link in the sidebar.

Name	Last commit	Last update
Auxiliary	issue #211	3 years ago
Infrastructure	Added	2 years ago
Tools	[ISSUE #208] -- MM tools updates	5 months ago
doc	report Fluctuations update issue #259	4 years ago
http	Issue #181: sources for backend	4 years ago
performance	Resolved merge conflicts, thor cannot find H5. Tha...	4 years ago
scripts	[ISSUE #225] hacked problem possibly crashing te...	4 months ago
src	[ISSUE #283]	2 weeks ago
.cvsignore	*** empty log message ***	6 years ago

Issue first

- I have an idea
- There is a bug
- I want to propose a new feature

Issue first

- I have an idea
- There is a bug
- I want to propose a new feature

The screenshot shows the GitLab interface for the 'kinectWebserver' project. The left sidebar contains navigation links: Overview, Repository, Issues (28), List, Board, Labels, Milestones, and Merge Requests (0). The main content area is titled 'Issues' and shows a list of open issues. At the top right of the issues list, there is a green 'New issue' button highlighted with a red rectangle. Below the button are filters for 'Open' (28), 'Closed' (86), and 'All' (114). A search bar and a 'Created date' dropdown are also present. The list of issues includes:

- test failing** (#114) · opened a week ago by Alessandro C · updated a week ago
- Detections are made but not retained if tracking is turned off** (#113) · opened 3 months ago by Werner Kroneman · updated 3 months ago
- Integrate new cylinder-on-head tracking method into kinectWebserver** (#112) · opened 4 months ago by Werner Kroneman · updated 4 months ago
- Suspicious code in prune_and_undistort** · updated 1

Issue at first

- Case of a bug --> the issue got ticket number #113

The screenshot shows the GitLab interface for the project 'kinectWebserver'. The top navigation bar includes links for Projects, Groups, Activity, Milestones, Snippets, and a search bar. The left sidebar contains navigation links for Overview, Repository, Issues (highlighted with 28 issues), Merge Requests (0), CI / CD, Registry, and Wiki. The main content area displays issue #113, titled 'Detections are made but not retained if tracking is turned off', which was opened 3 months ago by Werner Kroneman. The issue description states that 'df_json' is an empty list when 'global_tracking' is 0, despite detection steps being carried out. A code snippet is provided, showing a conditional statement that contradicts the comment. The right sidebar contains metadata for the issue, including Assignee (None), Milestone (None), Time tracking (No estimate or time spent), Due date (No due date), Labels (None), Confidentiality (Not confidential), and Lock issue.

GitLab Projects Groups Activity Milestones Snippets This project Search 75 32

kinectWebserver

Alessandro C > kinectWebserver > Issues #113

Open Opened 3 months ago by Werner Kroneman Close issue New issue

Detections are made but not retained if tracking is turned off

df_json is empty list if global_tracking is 0, despite the detection step being carried out and detection dots being visible on the merged image when global_detection is 1.

I suspect this code in pedestrian_vision is to blame:

```
## we need to produce this json just if there is no tracking. Otherwise it will be produced
if external_parameters.get_parameter("global_tracking",1) == 1:
    df_json = ret_merged['loc_df'].reset_index().to_dict(orient='records')
    print time_mark_pr.get_time_marked_string_prof("[ %s merge_and_show ] -
```

The comment contradicts the condition: it says it should only run if there is no tracking, yet the if-statement checks for a 1.

Not a high-priority issue on my side, though it was a bit confusing since I saw detection dots but no entries in the JSON.

Todo Add to

Assignee No assignee - assign yourself

Milestone None

Time tracking No estimate or time spent

Due date No due date

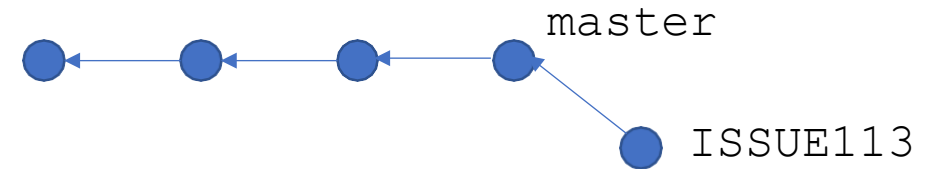
Labels None

Confidentiality Not confidential

Lock issue

From issue to code

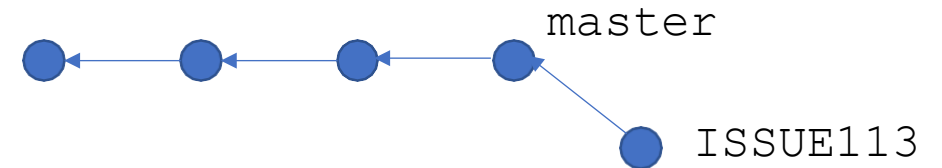
- If Ivan is in charge of fixing #113:
 - [Clones the code – if he does not have already]
 - Branches master -> e.g. into branch ISSUE113 (for trackability)
 - `git checkout -b ISSUE113`



From issue to code

- If Ivan is in charge of fixing #113:

- [Clones the code – if he does not have already]
- Branches master -> e.g. into branch ISSUE113 (for trackability)
- `git checkout -b ISSUE113`

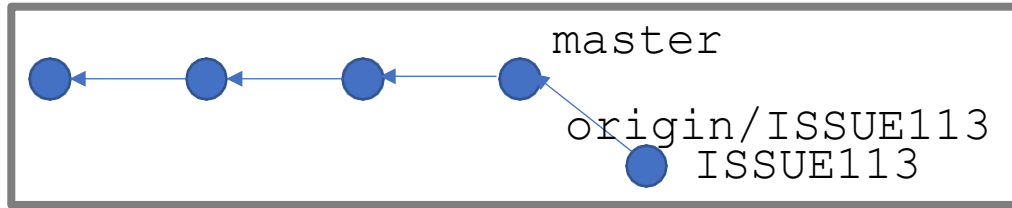


- Corrects bugs (& make regression tests)
- `git add code tests`
- `git commit -m '[ISSUE #113] - how code and tests have been changed'`
- `git push origin ISSUE113`

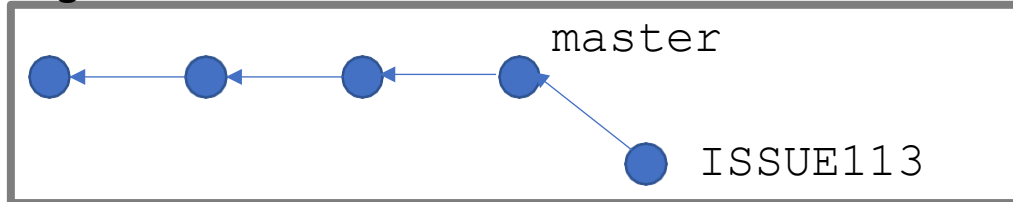
From issue to code

- `git commit -m '[ISSUE #113] - how code and tests have been changed'`
- `git push origin ISSUE113`

Ivan



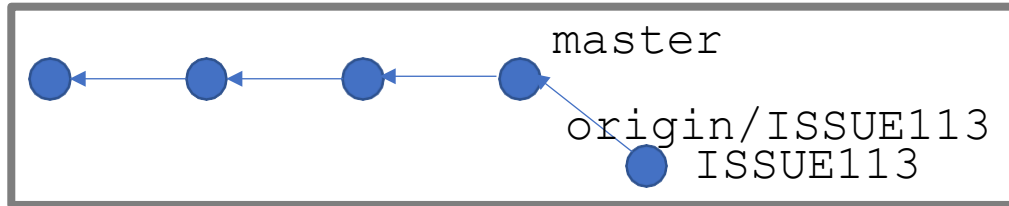
Origin



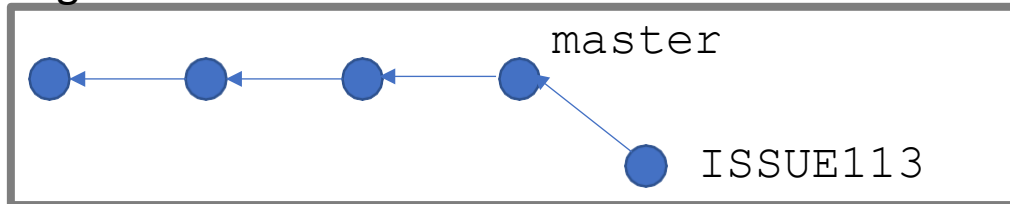
From issue to code

- `git commit -m '[ISSUE #113] - how code and tests have been changed'`
- `git push origin ISSUE113`

Ivan



Origin

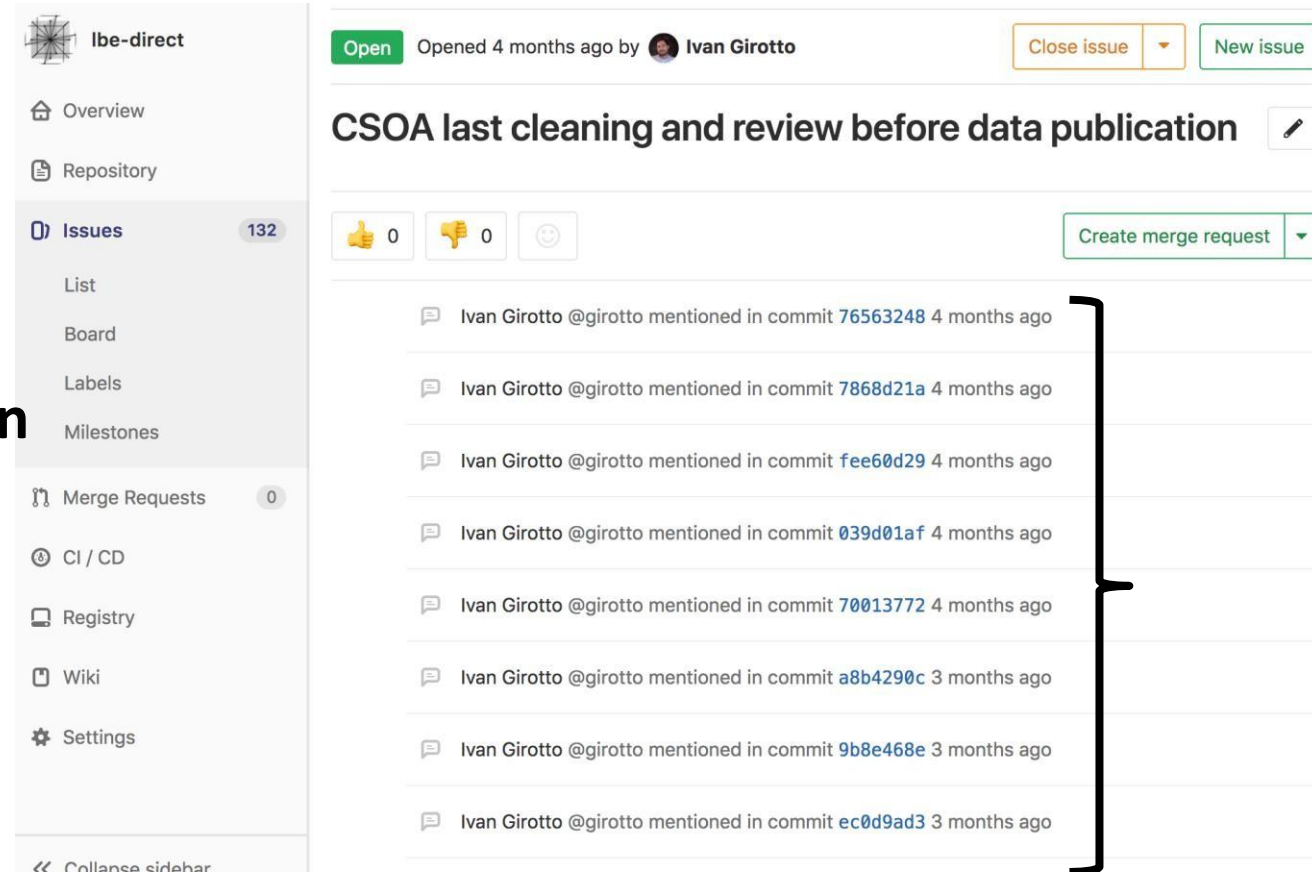


The screenshot shows the GitHub interface for the 'ibe-direct' repository. The left sidebar contains navigation links: Overview, Repository, Issues (132), List, Board, Labels, Milestones, Merge Requests (0), CI / CD, Registry, Wiki, and Settings. The main content area displays an issue titled 'CSOA last cleaning and review before data publication' opened 4 months ago by Ivan Girotto. The issue has 0 thumbs up, 0 thumbs down, and a 'Create merge request' button. The issue history shows several mentions in commits by Ivan Girotto, including commit 76563248, 7868d21a, fee60d29, 039d01af, 70013772, a8b4290c, 9b8e468e, and ec0d9ad3.

From issue to code

- `git commit -m '[ISSUE #113] - how code and tests have been changed'`
- `git push origin ISSUE113`

ALWAYS DO specify the issue number in the commit message. This will **link** each **contribution** to the related **motivation**



The screenshot displays the GitHub interface for the repository 'lbe-direct'. The left sidebar shows navigation options: Overview, Repository, Issues (132), List, Board, Labels, Milestones, Merge Requests (0), CI / CD, Registry, Wiki, and Settings. The main content area shows an issue titled 'CSOA last cleaning and review before data publication' opened 4 months ago by Ivan Girotto. Below the issue title, there are thumbs up (0) and thumbs down (0) buttons, and a 'Create merge request' button. A list of commit messages is shown, each mentioning the issue. A large black bracket on the right side of the commit list indicates that these commits are linked to the issue.

lbe-direct

Overview

Repository

Issues 132

List

Board

Labels

Milestones

Merge Requests 0

CI / CD

Registry

Wiki

Settings

Open Opened 4 months ago by Ivan Girotto Close issue New issue

CSOA last cleaning and review before data publication

0 0

Create merge request

Ivan Girotto @girotto mentioned in commit 76563248 4 months ago

Ivan Girotto @girotto mentioned in commit 7868d21a 4 months ago

Ivan Girotto @girotto mentioned in commit fee60d29 4 months ago

Ivan Girotto @girotto mentioned in commit 039d01af 4 months ago

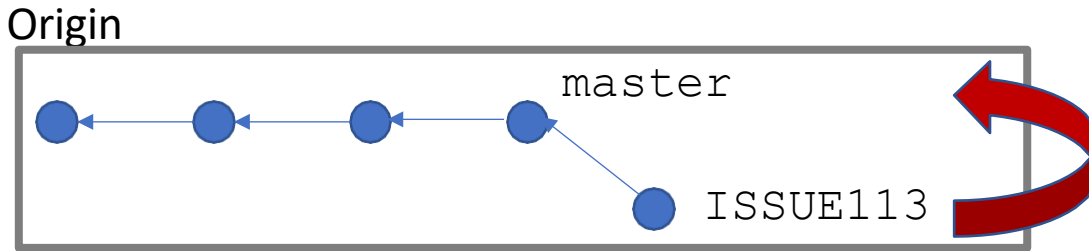
Ivan Girotto @girotto mentioned in commit 70013772 4 months ago

Ivan Girotto @girotto mentioned in commit a8b4290c 3 months ago

Ivan Girotto @girotto mentioned in commit 9b8e468e 3 months ago

Ivan Girotto @girotto mentioned in commit ec0d9ad3 3 months ago

How to contribute to master?

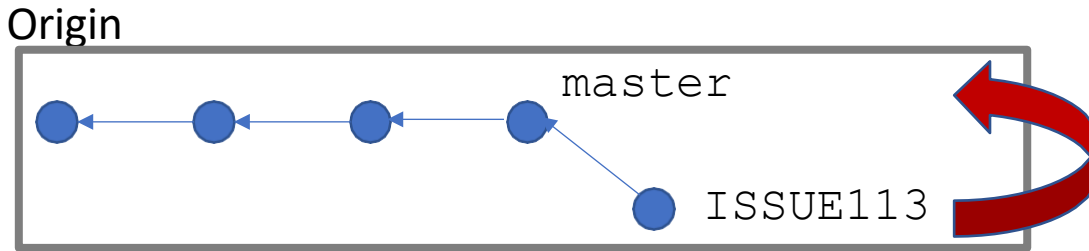


```
git push origin ISSUE113
```

master is a privileged branch.

Developers must not push directly! (generally prohibited)

How to contribute to master?



```
git push origin ISSUE113
```

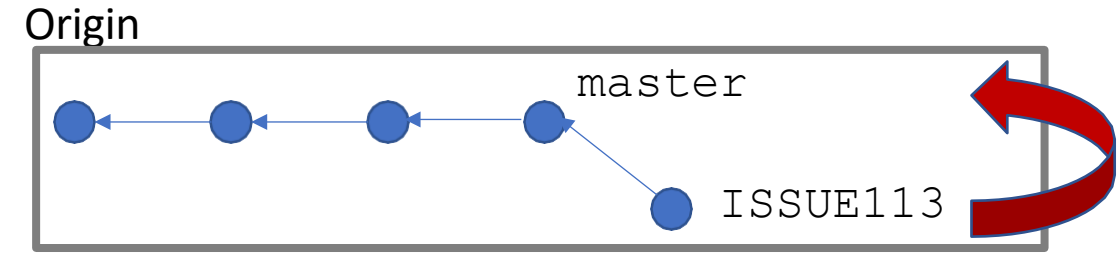
master is a privileged branch.

Developers must not push directly! (generally prohibited)

- **Required: code review from authoritative parties**
- Ask authority to **merge** contribution through merge request

Code review merge request

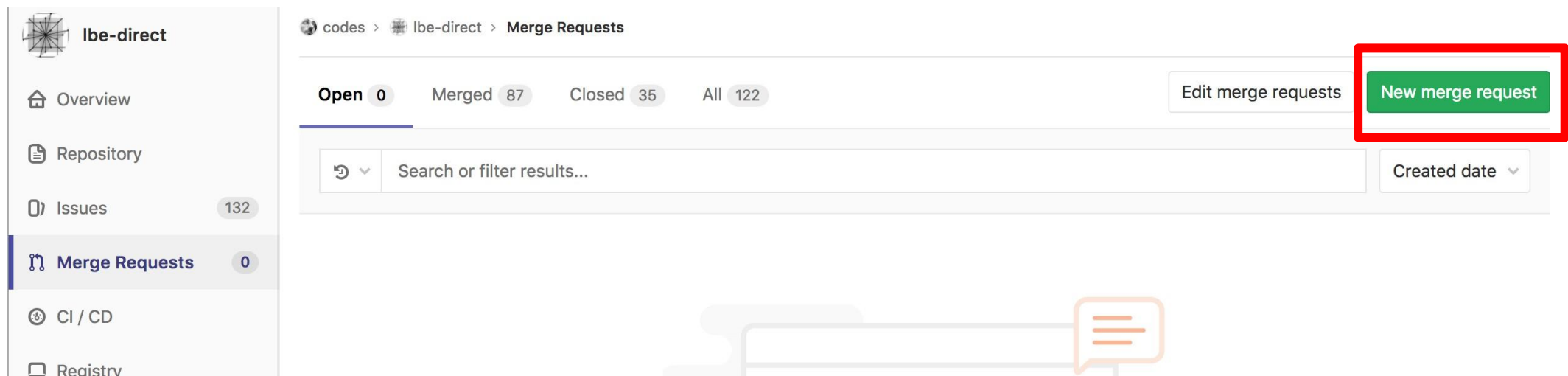
```
git push origin ISSUE113
```



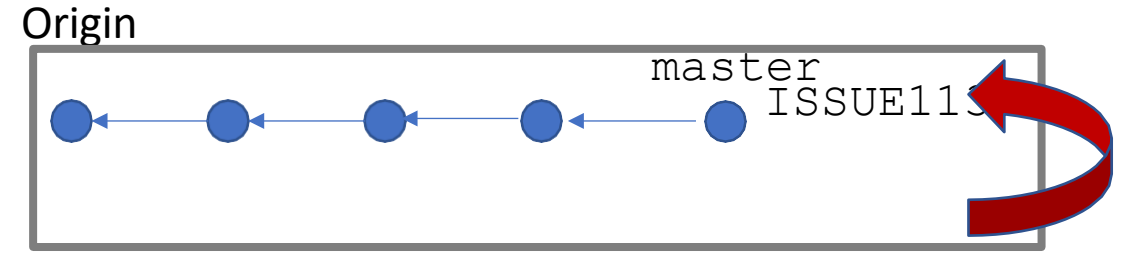
master is a privileged branch.

Developers must not push directly! (generally prohibited)

- **Required: code review from authoritative parties**
- Ask authority to **merge** contribution **through merge request**



Code review merge request



ftmake

Overview

Repository

Issues29

Merge Requests0

CI / CD

Wiki

Settings

Merged

Opened a week ago by Gianluca Di Staso

Edit

Close merge request

Issue 102

Modification to make the `dsmc_indirect` code run under ftmake.

Request to merge **ISSUE_102** into **master**

✓

Pipeline [#13356](#) passed for [e99181e7](#).

✓✓

✓

Merged by Federico Toschi 2 days ago

Revert

Cherry-pick

The changes were merged into **master**

You can remove source branch now

Remove Source Branch

👍 0

👎 0

😊

Discussion0

Commits4

Pipelines4

Changes4

🔗

Gianluca Di Staso @gianluca added 1 commit 4 days ago

- [e99181e7](#) - [ISSUE #102] -- Add new names of cartesius

[Compare with previous version](#)

Todo

Add todo

Assignee

Federico Toschi @toschi

Milestone

None

Time tracking

No estimate or time spent

Labels

None

Lock merge request

Unlocked

2 participants



Notifications

×

Reference: toschi/ftmake!71



Code review merge request

Code must be reviewed
before merging by an authority

Discussion 0Commits 4Pipelines 4Changes 4

Changes between latest version and master

Showing 4 changed files with 22 additions and 1 deletion

Hide whitespace changesInlineSide-by-side

CMakeLists.txt

@@ -182,7 +182,7 @@

include(\${FTMAKE_SOURCE_DIR}/cmake/aux.cmake)

#####

182#####

183set(CMAKE_MODULE_PATH \${CMAKE_MODULE_P

ATH} "\${FTMAKE_SOURCE_DIR}/cmake")

184set(CMAKE_MODULE_PATH \${CMAKE_MODULE_P

ATH} "\${PROJECT_SOURCE_DIR}/cmake")

185- #message(STATUS \${PROJECT_SOURCE_DIR})

186#####

187#####

188# Here I invoke the loading of machine

s specific hooks

...@@ -769,6 +769,11 @@

ADD_EXECUTABLE(\${TARGET_NAME}

main.F90)

769SET_TARGET_PROPERTIES(\${TARGET_NAME} P

ROPERTIES LINKER_LANGUAGE Fortran)

770endif(EXISTS "\${PROJECT_SOURCE_DIR}/ma

in.F90")

@@ -182,7 +182,7 @@

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#####

182#####

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Todo

Assignee

Federico Toschi

@toschi

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Reference: tosch/fttr

Structured collaboration

- Git for version control
- Developers contribute by (merging) separate branches
- Developers collaborate via **ISSUES**
- Interaction with «code authority» via merge **requests**
- Commit messages are expressive and related to ISSUES



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Now...

- Maintainer: “the merge request is in... but will it break the code?”
- Developer: “will this break the code?”

Nightly testing

- Did the the day worked ok?
 - 11PM – start tests suit current dev branches
 - 8AM – see problems; fix, else merge.
- Issues?

Aims

- Understand quickly if code (each branch) pass tests
- Merge frequently; avoid drifts
- Test before merging and merge
- Tests run quickly ($O(10\text{min})$)

Also

- Deploy frequently – “agile” dev-user cycle